KARL HILL

STAFF AEROSPACE SOFTWARE ENGINEER

CROSS-PROGRAM TECHNICAL LEADERSHIP | MISSION SOFTWARE & PLATFORM ENGINEERING

Washington, DC · (202) 599-1442 · karlhillx@gmail.com

linkedin.com/in/khill · github.com/karlhillx · karlhill.com

SUMMARY

Staff Aerospace Software Engineer and technical leader with 25+ years delivering mission-critical software across national security, aerospace, NASA, and enterprise environments. Leads cross-program engineering across multiple teams, shared repositories, partner organizations, and deployment environments while remaining hands-on in Python, distributed systems, integration, and developer tooling. Builds the engineering systems - architecture, CI/CD, DevSecOps, testing, release governance, and standards - that enable teams to deliver reliable software at scale.

LEADERSHIP & ENGINEERING SCOPE

Cross-Program Technical Leadership | Software Architecture | Platform Engineering & Developer Experience | Distributed Systems & Integration | DevSecOps & CI/CD | Engineer Development | Agile Delivery

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 — PRESENT

Jacobs National Security · BlackLynx · Chantilly, VA

- Provide cross-program technical leadership across multiple aerospace mission-software efforts, aligning internal and partner teams on engineering standards, shared interfaces, integration strategy, and release readiness.
- Lead technical execution for a core team of about 6 engineers across roughly 12 Python repositories and four deployment environments; sequence work, resolve cross-team dependencies, and drive integration and delivery.
- Design and develop mission software, distributed integrations, shared contracts, messaging capabilities, and service orchestration spanning RabbitMQ and ActiveMQ.
- Build and evolve platform engineering and DevSecOps capabilities including Bitbucket Pipelines, automated testing, dependency management, repository standards, quality/security gates, and release automation.
- Provide technical direction through architecture and design reviews, code review, interface decisions, repository governance, and resolution of cross-team implementation and integration issues.
- Onboarded and coached approximately six engineers through technical feedback, development guidance, engineering standards, and structured growth plans while leading cross-team SAFe Agile.

Lead Software Engineer

DEC 2017 — SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Architected and developed an AWS-based platform generating near-real-time flood and surface-water products from satellite and geospatial data; co-authored the peer-reviewed GeoHorizons publication describing the system.
- Helped rebuild NASA Earth Observatory, serving approximately 1.5 million monthly visitors; the team received a NASA Group Achievement Award.
- Delivered LAADS DAAC Find Data search, ordering, and near-real-time access, with GitLab CI/CD and Kubernetes-based web delivery alongside existing archive services.
- Led modernization of legacy scientific processing workflows into containerized services supported by automated CI/CD and Kubernetes deployment.
- Built an automated content-registry workflow that improved scientific data-collection efficiency by approximately 60%, and developed Ceph-based file and metadata services for large datasets.
- Led Agile technical delivery across engineers, scientists, operations teams, and program stakeholders while strengthening testing, code review, documentation, and production-readiness practices.

DETAILS

Washington, DC

(202) 599-1442

karlhillx@gmail.com

LINKS

LinkedIn
linkedin.com/in/khill

GitHub
github.com/karlhillx

Website
karlhill.com

AREAS OF EXPERTISE

- Cross-Program Technical Leadership
- Software Architecture
- Platform Engineering & Developer Experience
- Distributed Systems & Integration
- DevSecOps & CI/CD
- Engineer Development
- Agile Delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 — DEC 2017

InformedDNA · Washington, DC

- Designed and delivered a Laravel case-management platform connecting counseling workflows, documentation, and billing.
- Built CRM enhancements for customer lifecycle workflows and reporting, and tightened maintenance and security processes with operations.

Senior Software Engineer

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, DC

- Delivered SugarCRM solutions for more than 20 clients, including the Virginia Department of Transportation and Kastle Systems.
- Improved backlog management and delivery practices across client projects.

Earlier Engineering Experience

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal-level roles across travel, enterprise CRM, telecommunications, and managed security.
- At Sabre, built new PHP applications and added features to existing ones for large-scale travel systems.

AREAS OF EXPERTISE

Cross-Program Technical Leadership · Software Architecture · Platform Engineering & Developer Experience · Distributed Systems & Integration · DevSecOps & CI/CD · Engineer Development · Agile Delivery

PUBLICATIONS

Peer-reviewed research: A web-based high-resolution global water and flood mapping platform. Co-author, A web-based high-resolution global water and flood mapping platform, GeoHorizons (AGU / Geological Society of London), 2026. Software engineering for a NASA-supported system delivering near-real-time, high-resolution satellite-derived water and flood products worldwide. Author contribution (CRediT): Software (Equal); Writing – review & editing (Equal). doi:10.1144/gh2025-7

EDUCATION

Computer Science coursework, University of Maryland, Baltimore County

Associate of Arts, General Studies — Computer Science Curriculum, Howard Community College

Project Management Certificate, Rutgers University

CERTIFICATIONS

SAFe Agilist , Scaled Agile

Professional Scrum Master II , Scrum.org

Professional Scrum Product Owner I , Scrum.org

INDEPENDENT PRODUCTS

The Dry Standard — Production non-alcoholic drink catalog & comparison engine.

OPEN SOURCE

bb-run — Run Bitbucket Pipelines locally.

testrisk — Rank the highest-value Python test gaps from coverage, AST, and git.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, Bash

Cloud & Containers: AWS, Docker, Kubernetes, Helm

CI/CD & Developer Tooling: Git, Bitbucket Pipelines, GitLab CI, GitHub Actions, release automation, security scanning

Python Tooling: uv, Ruff, ty, pytest, pre-commit — dependency management, linting, type checking, testing, and automated quality gates

Integration: REST/OpenAPI, RabbitMQ, ActiveMQ, distributed systems

Data: PostgreSQL, MySQL, SQL Server, Elasticsearch/OpenSearch, Redis